

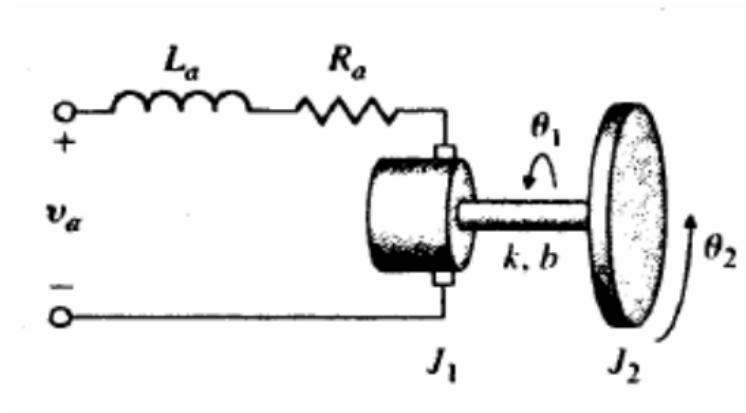
# Homework 4

EE 999 — Introduction to Control Systems

Patrick Gordon • October 1, 2026

*Due Wednesday, October 7, 2026*

---

**Problem 1.A** — Equations of MotionFigure 1: DC motor driving a load inertia  $J_2$ 

Electrical side:

$$v_a(t) = v_L(t) + v_R(t) + v_{J_1}(t)$$

$$v_L(t) = L_a \dot{i}_a(t)$$

$$v_R(t) = R_a i_a(t)$$

$$v_m(t) = K_e \dot{\theta}_1(t)$$

Rotor:

$$M_c \ddot{\theta}_1(t) = T_{net.J_1}(t) = T_e(t) - T_B(t) - T_k(t) - T_b(t)$$

$$T_e(t) = K_t i_a(t)$$

$$T_B(t) = B \dot{\theta}_1(t)$$

$$T_k(t) = k(\theta_1 - \theta_2)$$

$$T_b(t) = b(\dot{\theta}_1 - \dot{\theta}_2)$$

Load:

$$M_{load} \ddot{\theta}_2(t) = T_{net.J_2}(t) = T_b(t) + T_k(t)$$

## Standard Forms

$$v_a(t) = L_a \dot{i}_a(t) + R_a i_a(t) + K_e \dot{\theta}_1(t)$$

$$\begin{aligned} M_c \ddot{\theta}_1(t) &= K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2) \\ M_c \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + b \dot{\theta}_1 + k \theta_1(t) &= K_t i_a(t) + k \theta_2 + b \dot{\theta}_2 \end{aligned}$$

$$\begin{aligned} M_{load} \ddot{\theta}_2(t) &= k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t)) \\ M_{load} \ddot{\theta}_2(t) + b \dot{\theta}_2(t) + k \theta_2(t) &= k \theta_1(t) + b \dot{\theta}_1(t) \end{aligned}$$